

Recombinant DNA and Genetic Engineering

Recombinant DNA

Gene splicing → Genetic Engineering

Defination:

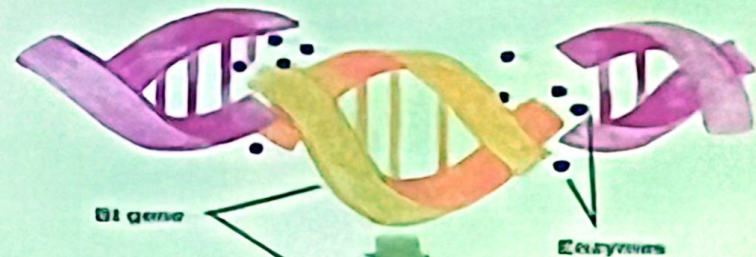
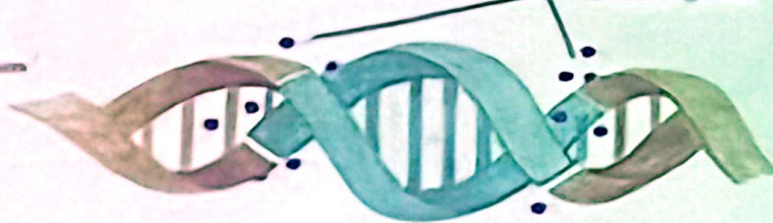
Is the artificial creation of unique DNA segments, the parts of which are derived from different biological sources.

Old genes with
foreign genes removed
superficial association



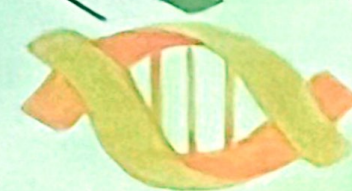
Enzymes are used
to move genes

Gene



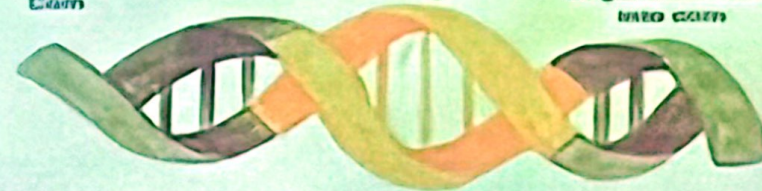
Old gene

Enzymes

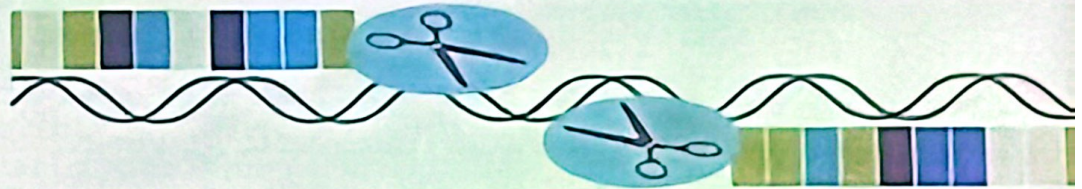


Gene

Old gene inserted
into gene



Restriction Endonucleases

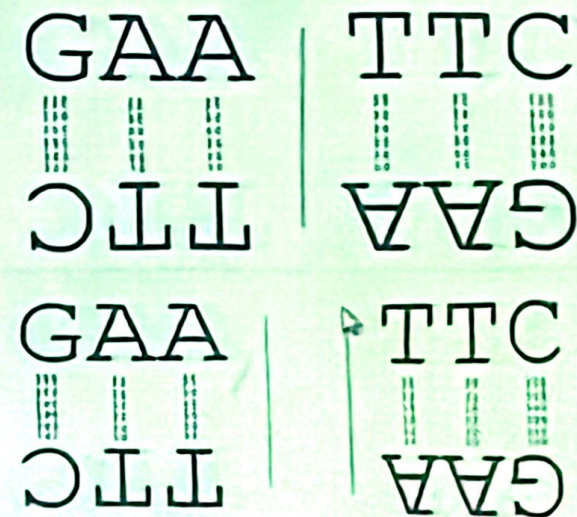


- The cornerstone of recombinant DNA technology
- These enzymes, isolated from bacteria,
- received their name because they restrict viral infection by degrading the invading nucleic acid.
- Two classes of restriction endonucleases: class I and II
- More than 80 enzymes of both types are now recognized

Class I restriction Endonucleases

Recognize a specific nucleotide sequence and produce a nonspecific double stranded cut in the DNA molecule close to that sequence.

Blunt ends



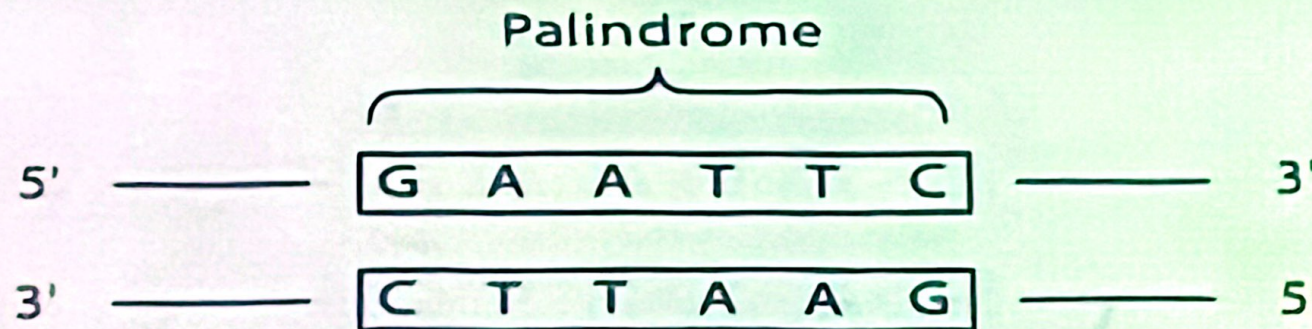
Class II restriction Endonucleases

Recognize a specific sequence and produce a double-stranded cut precisely within the sequence.

The recognition site for the type II enzymes is palindromic

Some of the type II enzymes produce two DNA fragments with single strand complementary tails.

Ex. The endonuclease EcoRI, obtained from *E. coli*, has the following palindromic recognition site and points of cleavage:

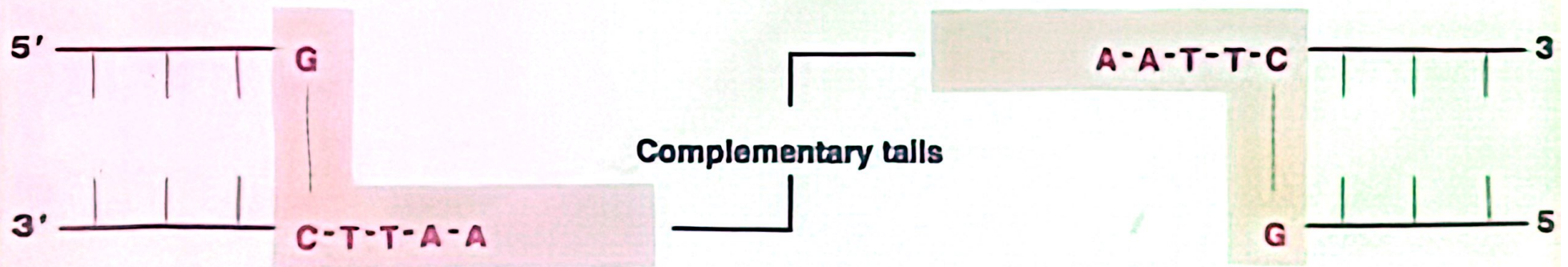


MCAT-Review.org

Recognition site

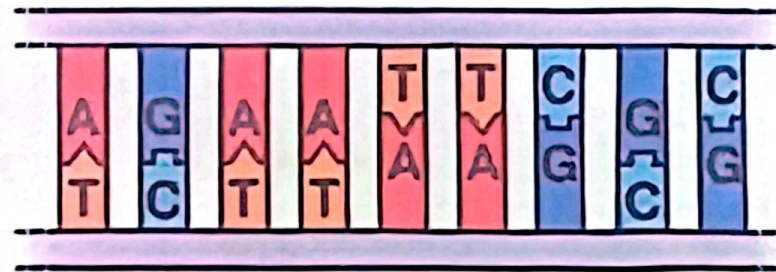


Treatment with *Eco*RI

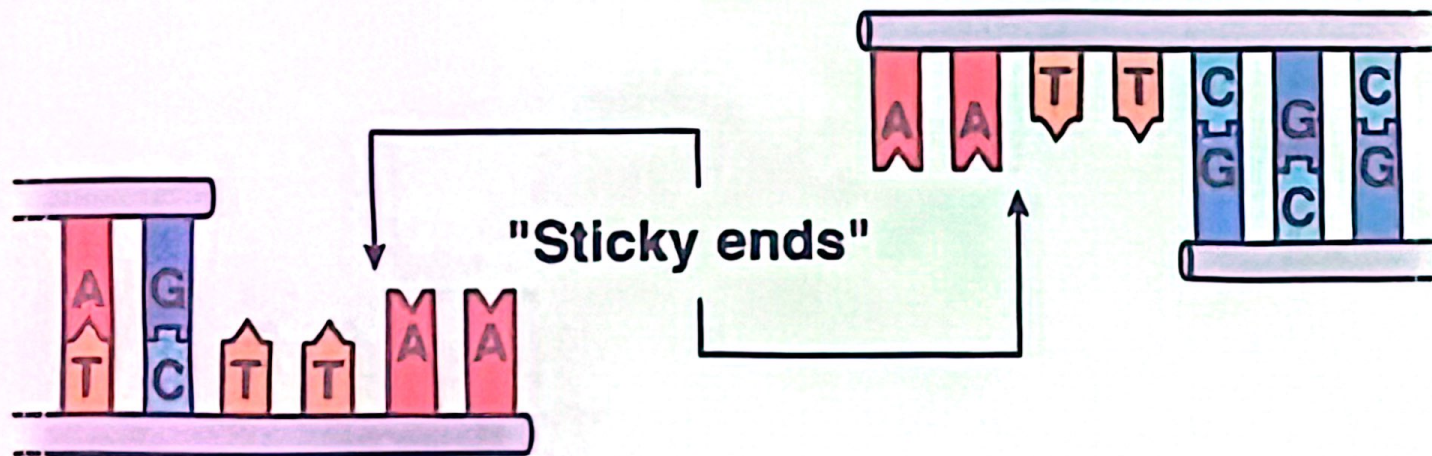


DNA

DNA
duplex



↓
restriction
enzyme

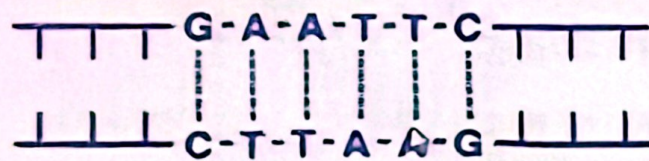


Class II restriction Endonucleases

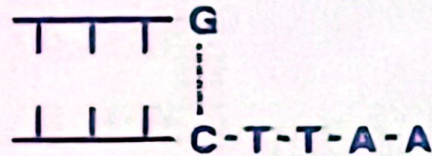
The complementation tails are said to be "sticky" because under **hybridization** conditions, they will reanneal (complement, make hydrogen bonds) with each other

If the DNA from two sources share the same palindromic recognition sites in their DNA sequence, then both will contain complementary single strand tails if enzymatically treated.

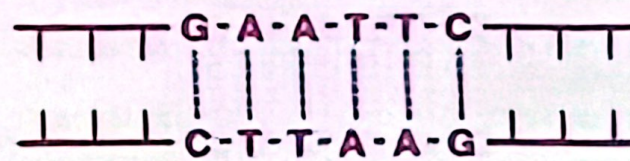
When placed together under proper condition, recombinant DNA molecules can be created.



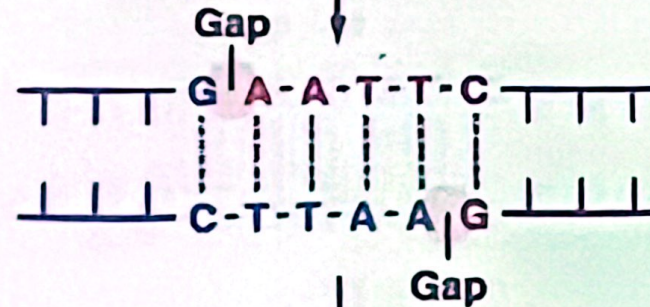
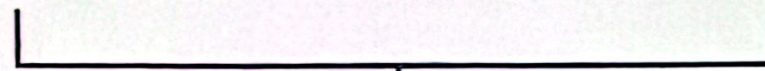
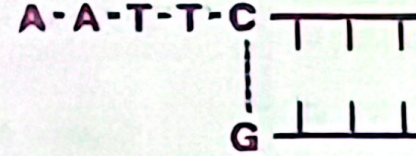
Cleavage with *Eco*RI



Fragments with complementary tails

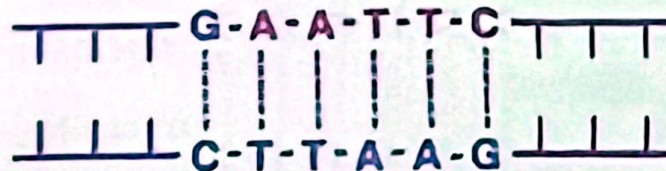


Cleavage with *Eco*RI



Annealing allows recombinant DNA molecules to form by complementary base pairing. The two strands are not covalently bonded as indicated by shaded gaps.

DNA ligase



DNA ligase seals the gaps, covalently bonding the two strands.

Recognition sequences of some restriction endonucleases.

Name	Recognition sequence	End after cleavage*		Source
Eco Ri	↓ □ GAATTC □ □ CTTAAG □ ↑	□ G □ CTTAAG	AATTC □ G □	E. coli containing drug resistance plasmid RI
Hind III	↓ □ AAGCTT □ □ TTCGAA □ ↑	□ A □ TTCGA	AGCTT □ A □	Hemophilus Influenzae, serotype D
Bam I	↓ □ GGATCC □ □ CCTAGG □ ↑	□ G □ CCTAG	GATCC □ G □	Bacillus amyloliquefaciens
Hae III	↓ □ GGCC □ □ CCGG □ ↑	□ GG □ CC	CC □ GG □	Hemophilus aegyptius

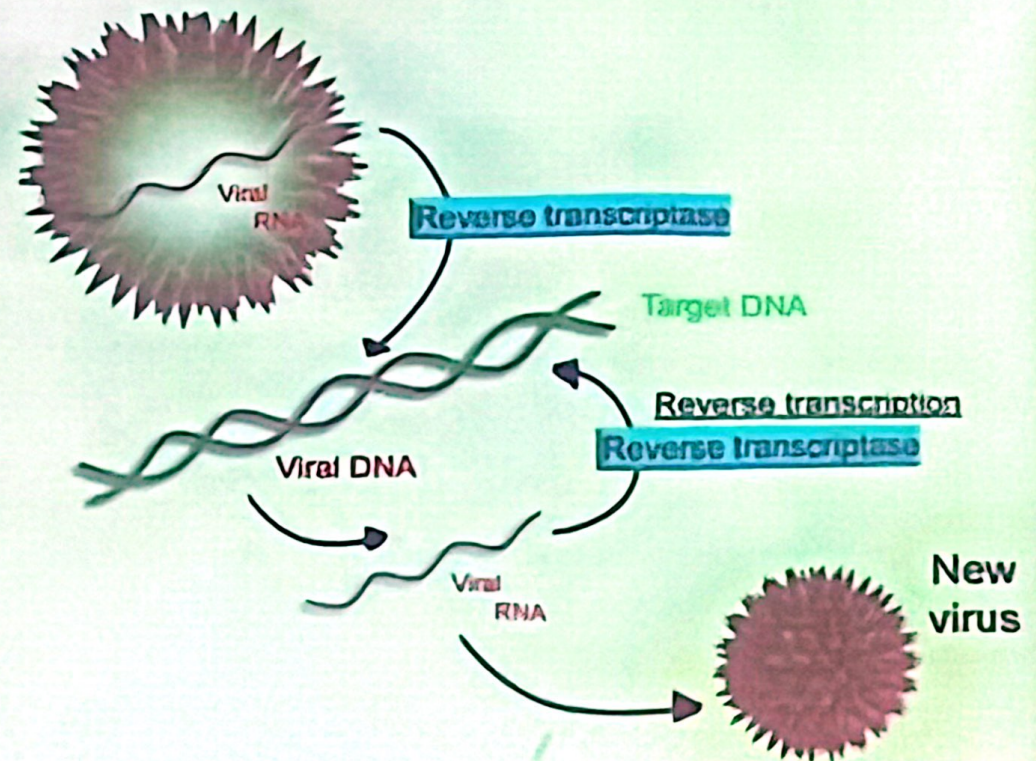
2. Reverse Transcriptase:

Very important enzyme which can transcribe DNA molecules "backwards" from RNA templates.

It was discovered in some viruses whose entire genome consists only of RNA.

Because RNA is not capable of self replication, the only way for these viruses to propagate is convert their

RNA
↓
Reverse transcriptase.
DNA



Reverse Transcriptase

- Reverse transcriptase is important in the recombinant DNA technology because it is easier to isolate RNA than DNA from cells

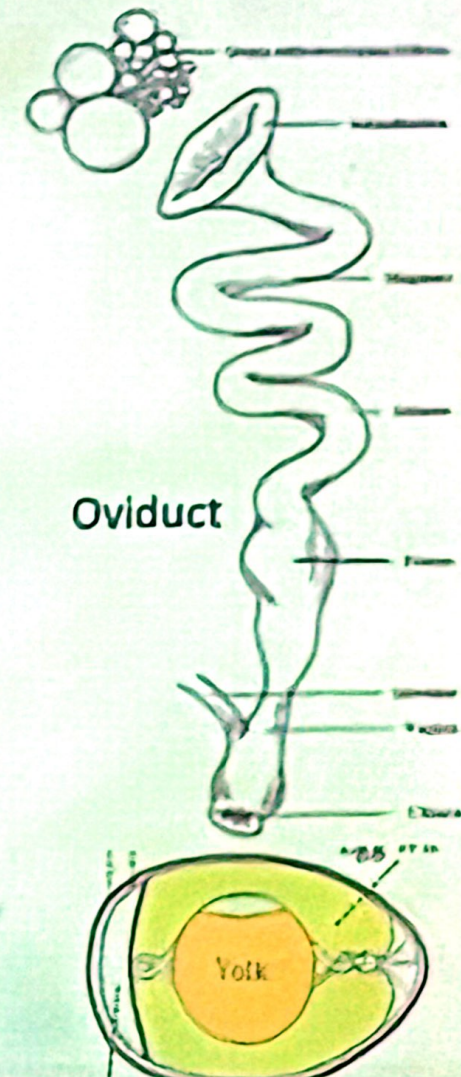
Example

Oviduct in chickens produce large quantities of ovalbumin and very little else.

It is easy to isolate large quantities of ovalbumin RNA from chicken oviduct cells.

↓
Reverse Transcriptase

↓
cDNA



5' **A-A-A-A** 3' mRNA
3' **T-T-T-T** 5' Oligo(dT) primer

- ① Incubate with reverse transcriptase to synthesize cDNA strand

Poly dT primer



mRNA/cDNA double stranded duplex

- ② When cDNA strand is completed, hydrolyze RNA strand

Hydrolysis of RNA by Alkali or RNaseH enzyme

Hairpin



- ③ Incubate with DNA polymerase to synthesize second DNA strand

Hairpin act as primer

S1 nuclease cuts loop



Digest hairpin by S1 nuclease

④



Double-stranded cDNA ready for insertion into appropriate cloning vector

Reverse Transcriptase

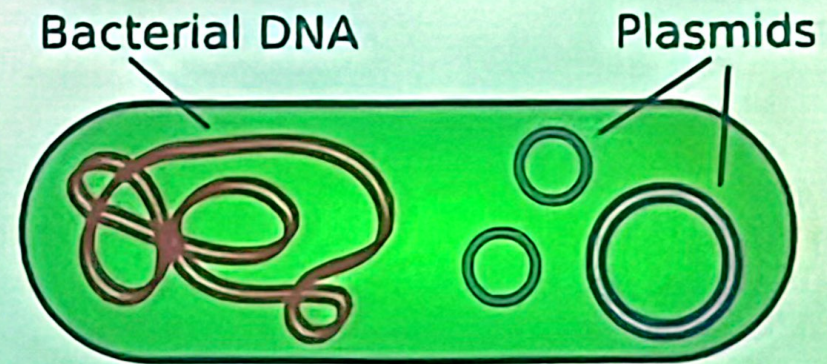
- The production of cDNA from mRNA
- Because many Eukaryotic mRNA have polyadenylated tail (Poly A tail) at their ends, a short poly dT tail can be annealed to this tail and serve as a primer for Reverse Transcriptase enzyme

Vectors

Are carriers of DNA

Types

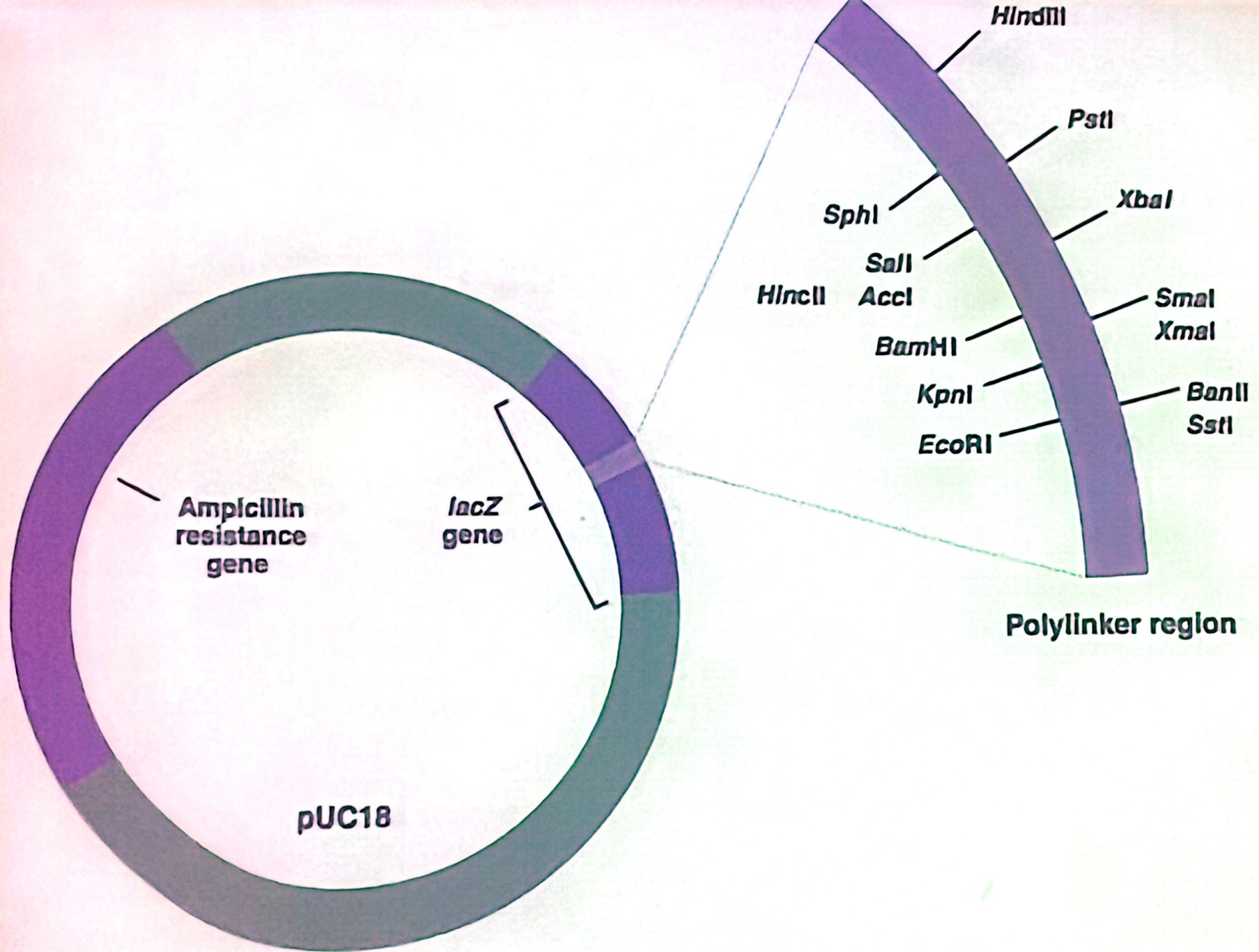
- **Plasmid**
- **Phage lambda that infect E. coli,**
- **DNA animal virus.**

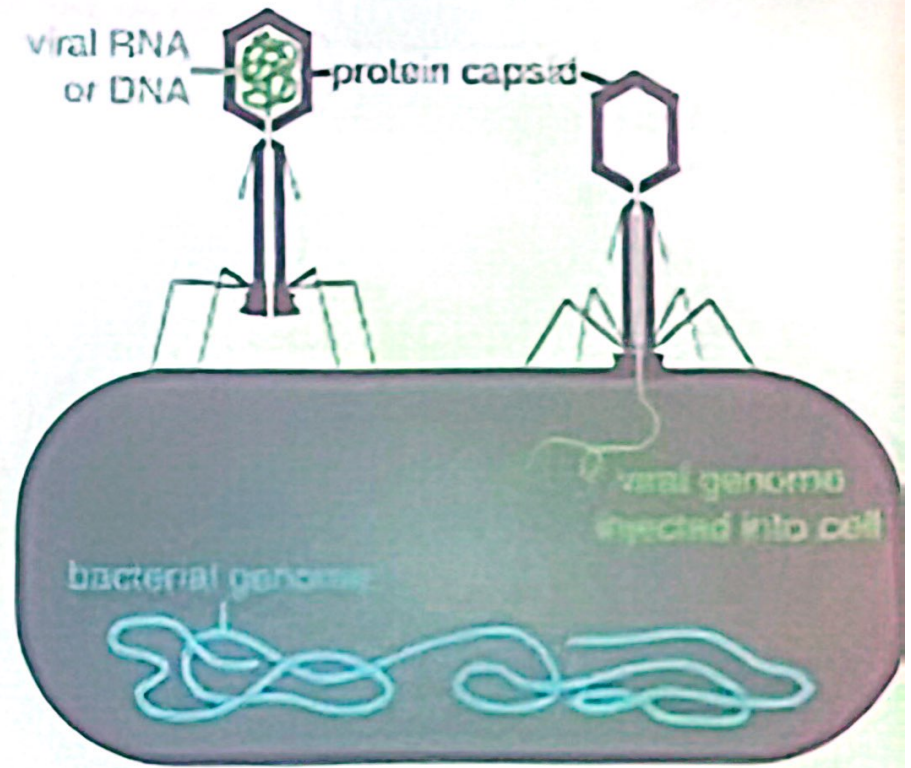
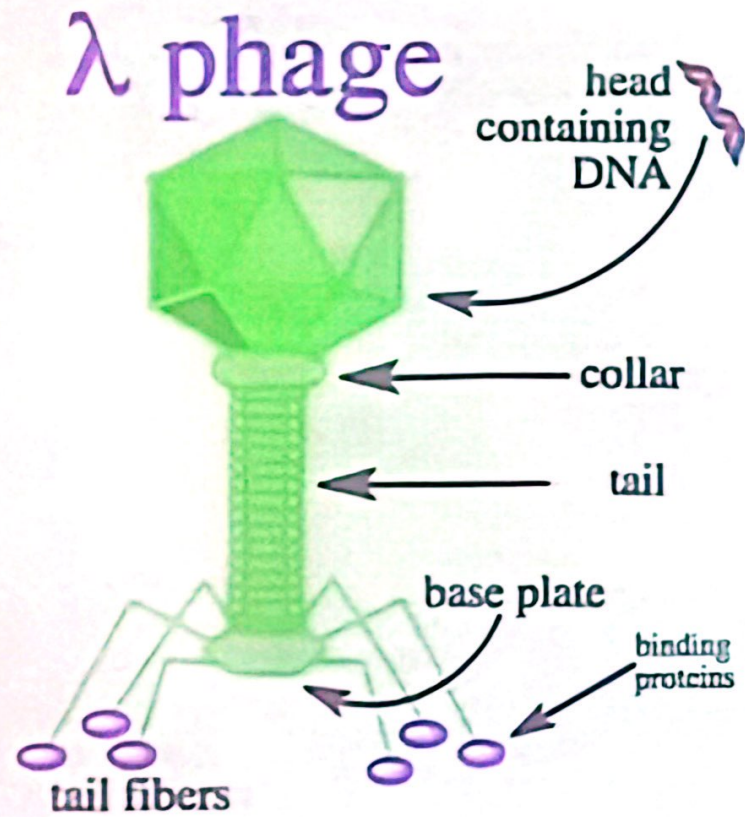


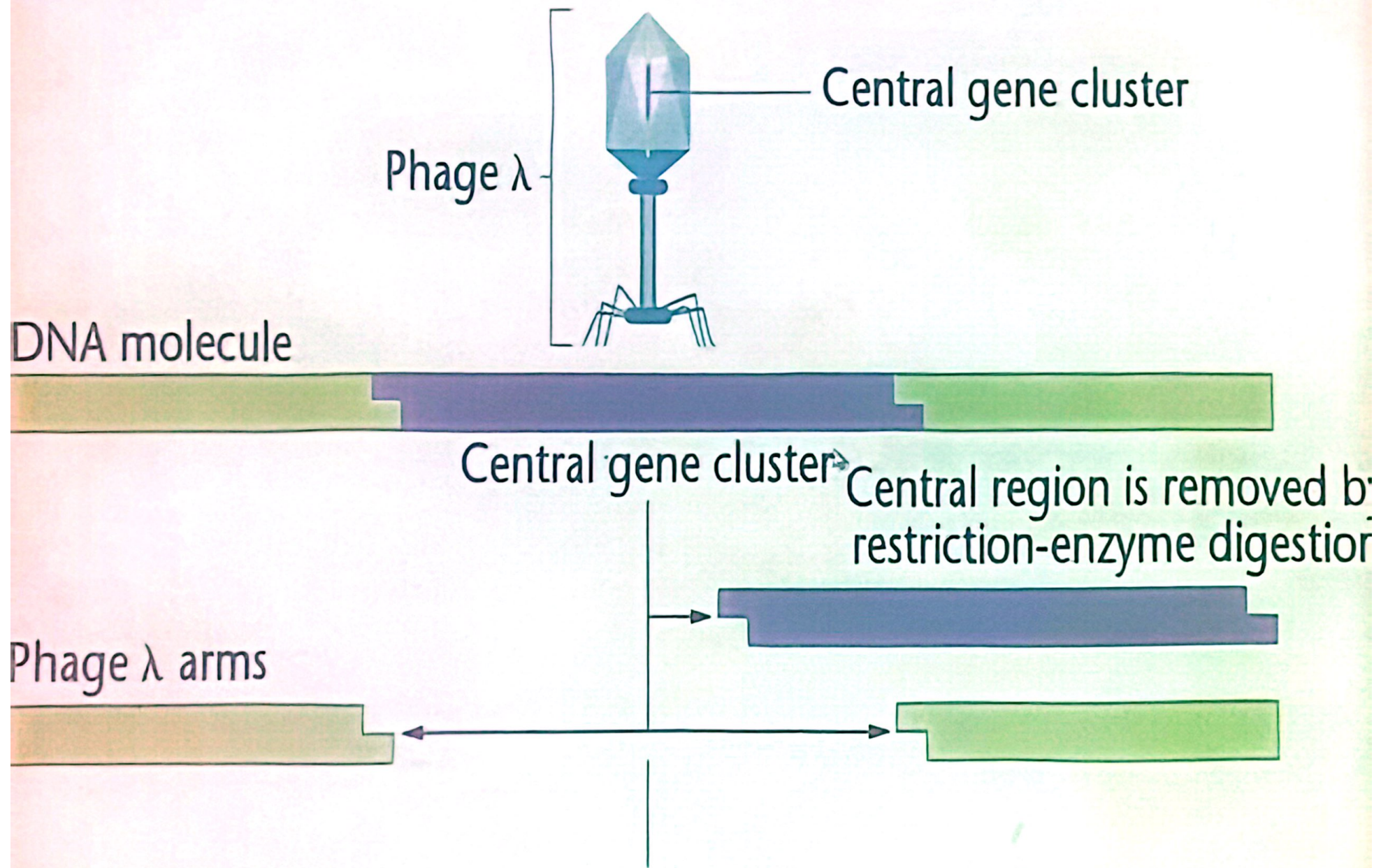
- **Plasmid**
- Are naturally occurring extrachromosomal double-stranded circular DNA molecules that replicate autonomously within bacterial cells.
- single plasmid may enter a host cell, many plasmids can increase their copy number in the host cell so that up to 1000 copies

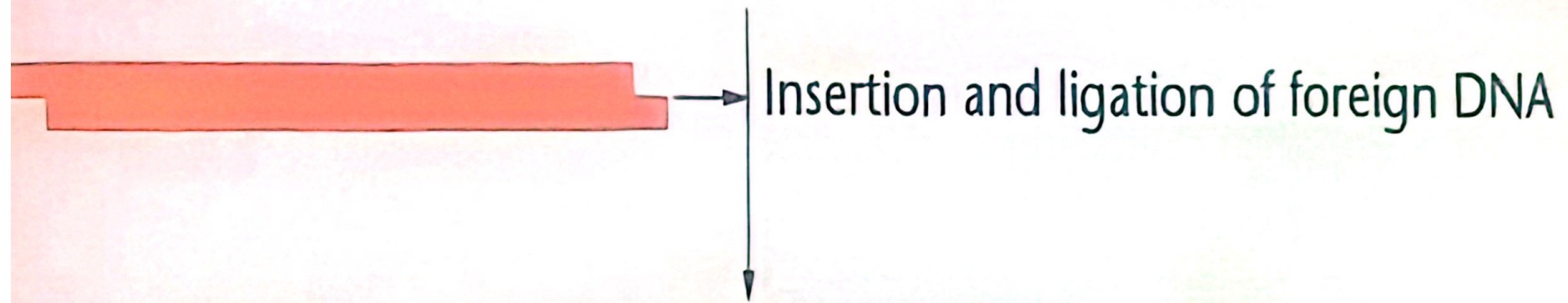
Plasmid

- Such plasmids allow more copies of cloned DNA to be produced.
- Plasmids used as a vector has been modified in the laboratory to
 - 1. Contain a limited number of restriction sites for the insertion of DNA fragments
 - 2. Marker genes that can be used to detect the presence of plasmids in host cells. As antibiotic resistance genes
- Many genetically engineered plasmid vectors are now available such plasmid is PUC18.







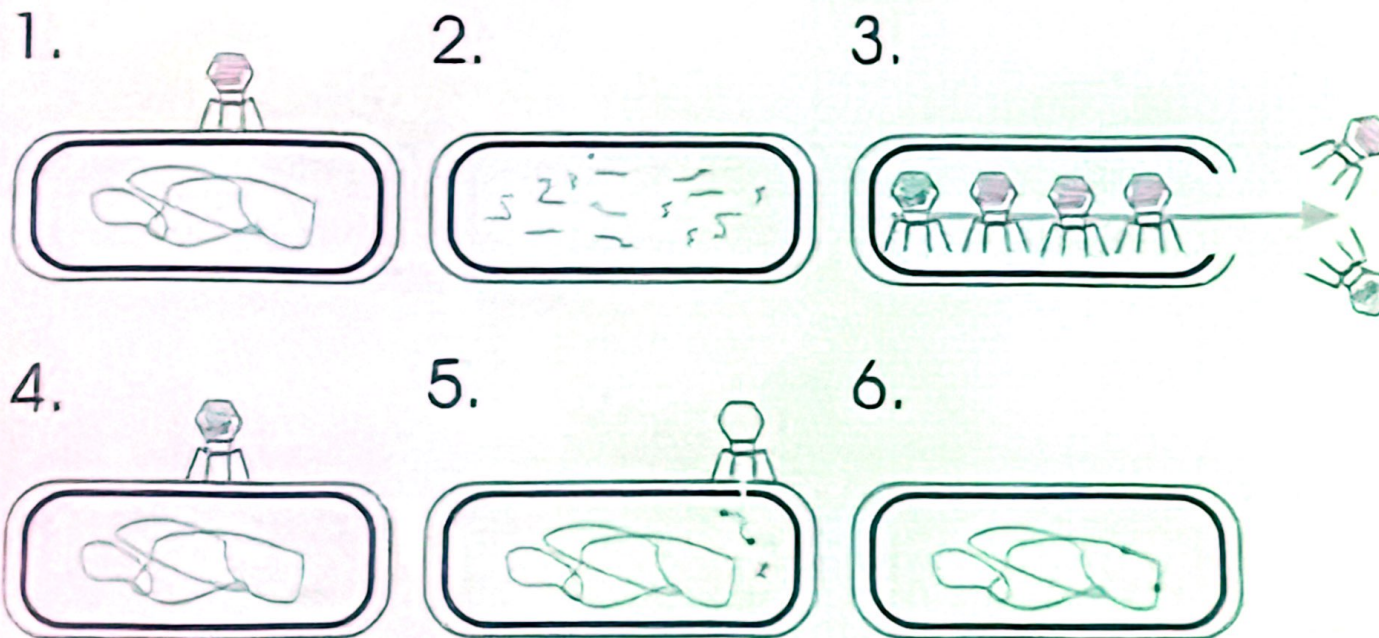
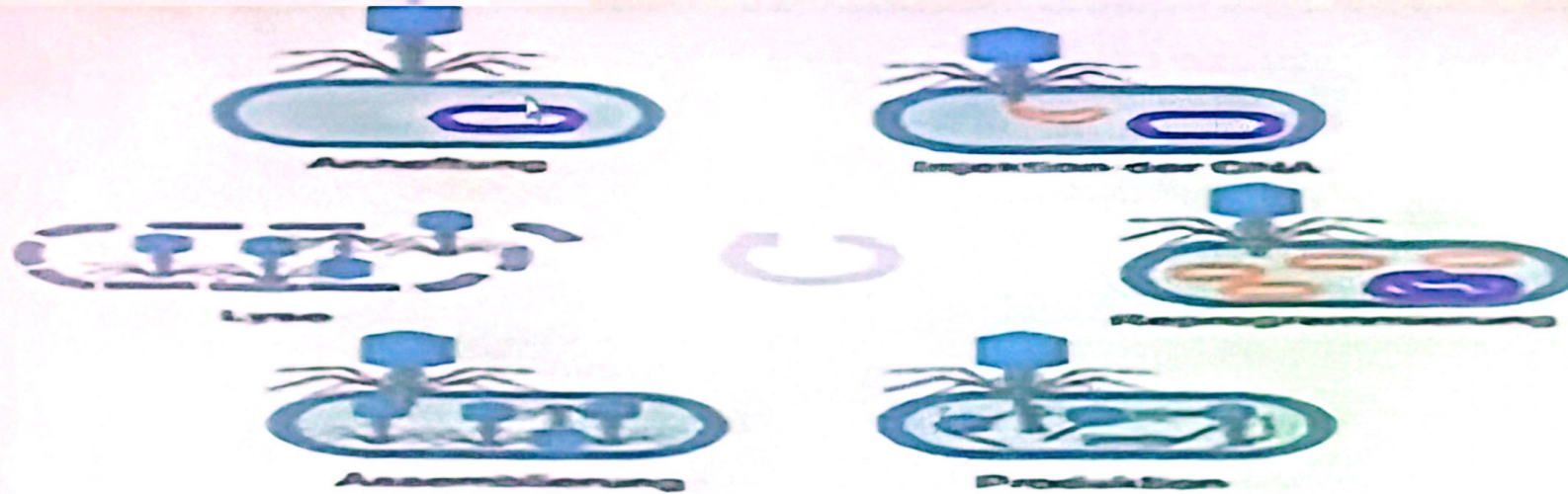


↓ *in vitro* Packaging



A recombinant viral particle able to infect a bacterial host cell carries a cloned foreign insert and replicates to form plaques

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■ Bacterial DNA ■ Viral DNA

Vectors: 2. Bacteriophage

- Phages are viruses that infect and kill bacterial cells.
- One of the most widely used of these vectors is *phage lambda*.

The middle-one-third of the lambda chromosome contains a cluster of dispensable genes, and this region can be replaced with foreign DNA without affecting the ability of the phage to infect cells

As they reproduce, the phages form plaques from which the cloned DNA can be recovered.

Advantage: Phage vectors are being able to carry much longer DNA fragments than plasmids

Steps of Recombinant DNA technology

The following are the general outlines (steps) of recombinant DNA technology:

- a. Obtaining DNA segments**
- b. Joining DNA Fragments to Vector DNA Molecules**
- c. Cloning Recombinant DNA**
- d. Production of Polypeptide from Cloned DNA**

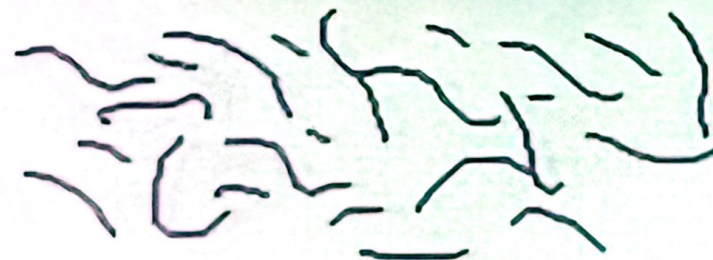
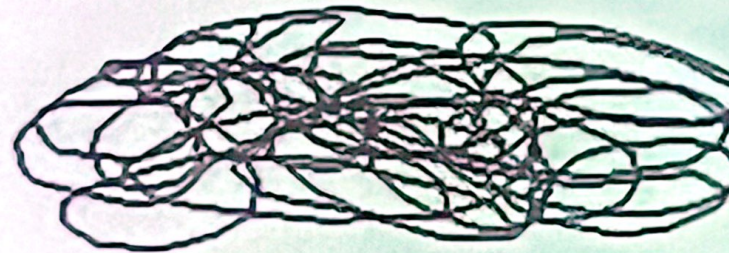
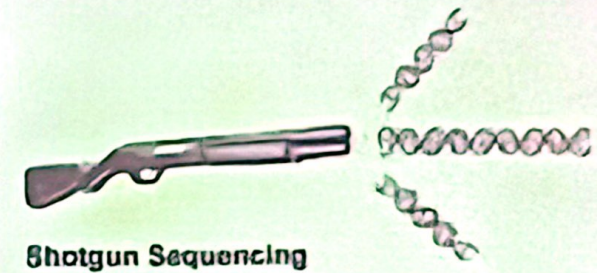
Steps of Recombinant DNA technology

a. Obtaining DNA segments

Two basic approaches are

1. Shot-gun experimentation

Cutting DNA into fragments randomly with restriction endonucleases. This results in a random DNA fragments, which can then be isolated and studied.



Steps of Recombinant DNA technology

2. Obtaining DNA segments specific for certain genes

Specific mRNA molecule is isolated



Reverse Transcriptase



cDNA

Example

Hemoglobin mRNA is nearly the only mRNA transcript produced at a certain stage of red blood cell development

Ovalbumin mRNA can be obtained from the uterine tissue of chickens.

Steps of Recombinant DNA technology

b. Joining DNA Fragments to Vector DNA Molecules:

A vector is a carrier of the inserted DNA fragment.

In order to be useful, the vector must have three properties:

- i. It can replicate.
- ii. Its presence can be detected.

Steps of Recombinant DNA technology

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In order to be useful, the vector must have three properties:

- i. It can replicate.
- ii. Its presence can be detected.
- iii. It is capable of invading a cell.

Joining a DNA fragment to a plasmid DNA by two methods:

1. if both had been enzymatically treated with a type II restriction endonuclease,

Both DNA's will have complementary single-stranded tails.

When placed together under proper conditions, these "sticky" ends or tails will reanneal, and DNA ligase is used to seal the open ends.

Steps of Recombinant DNA technology

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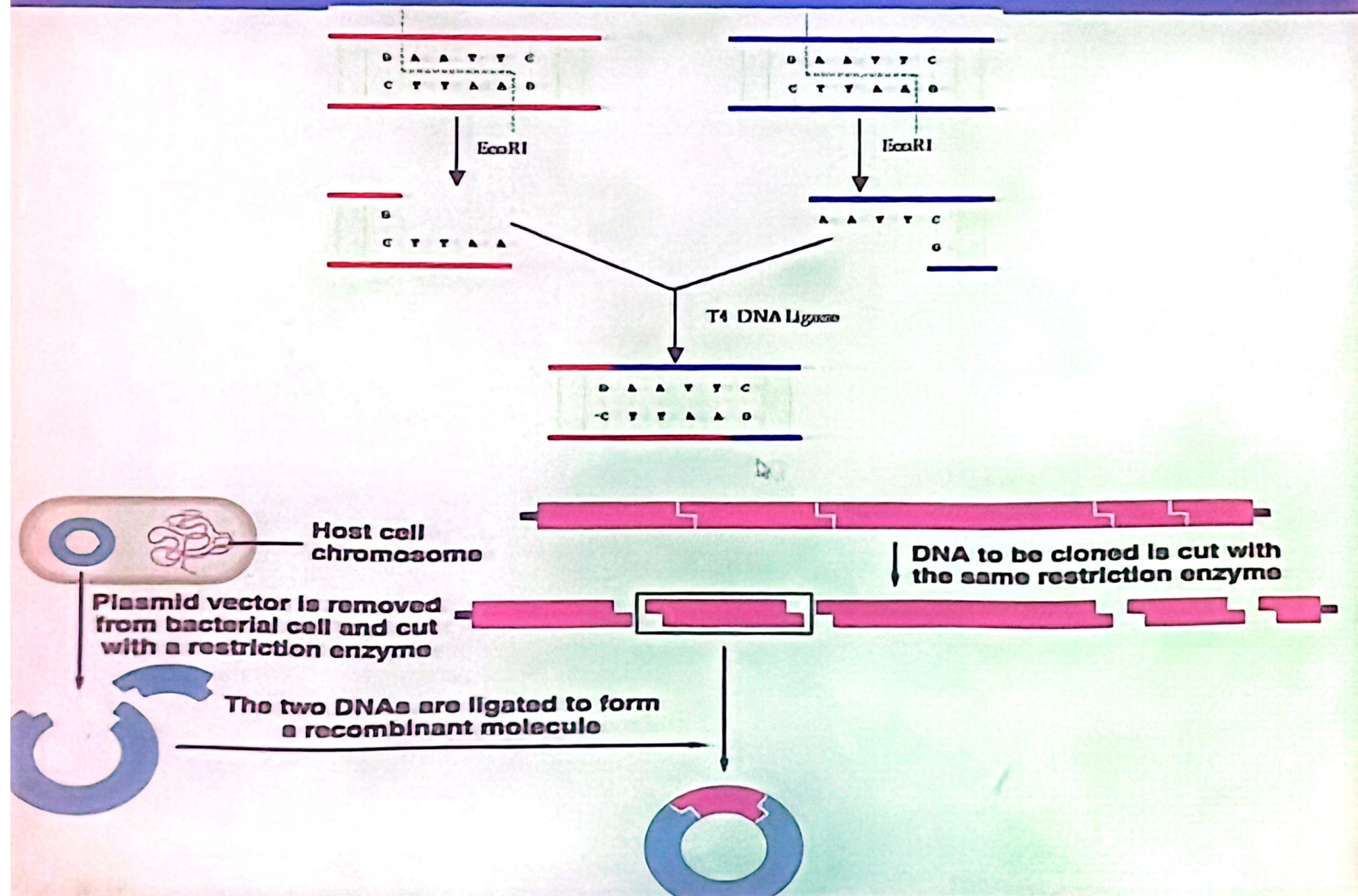
1. if both had been enzymatically treated with a type II restriction endonuclease,

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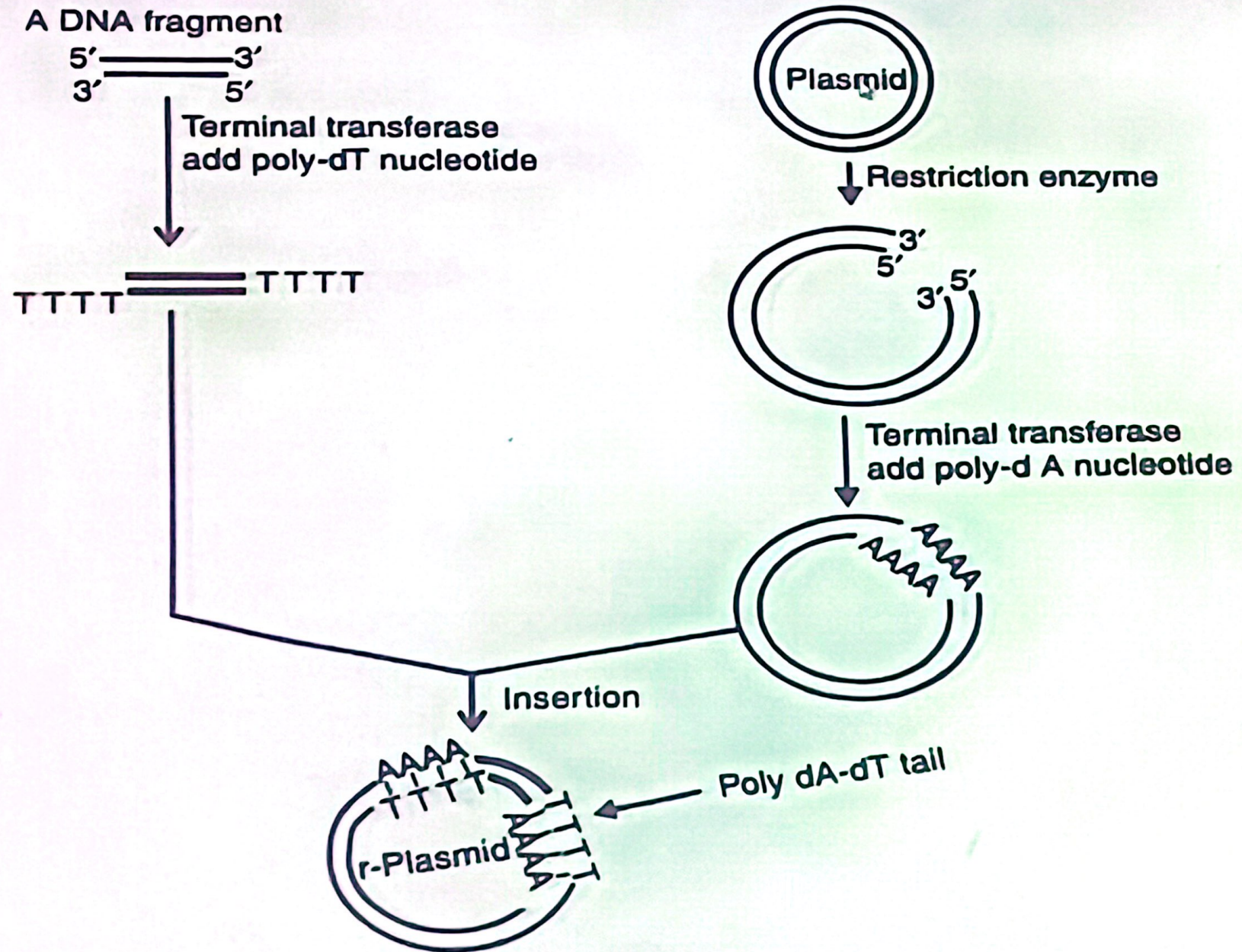
When placed together under proper conditions, these "sticky" ends or tails will reanneal, and DNA ligase is used to seal the open ends.

Thus, a hybrid plasmid or recombinant DNA is produced

Steps of Recombinant DNA technology



Steps of Recombinant DNA technology



Steps of Recombinant DNA technology

Joining a DNA fragment to a plasmid DNA by two methods:

2. Using of type I restriction endonucleases.

The double-stranded cut in DNA's to be joined are extended with single-stranded complementary ends.

Using an enzyme called *terminal deoxynucleotidyl transferase*

It add a tail of 50 to 200 adenine or thymine nucleotides.

These complementary tails are sticky and will anneal in the same way described in the previous methods .

Steps of Recombinant DNA technology

c. Cloning Recombinant DNA:

Making many copies, or clones.

Cloning in E. coli host cells:

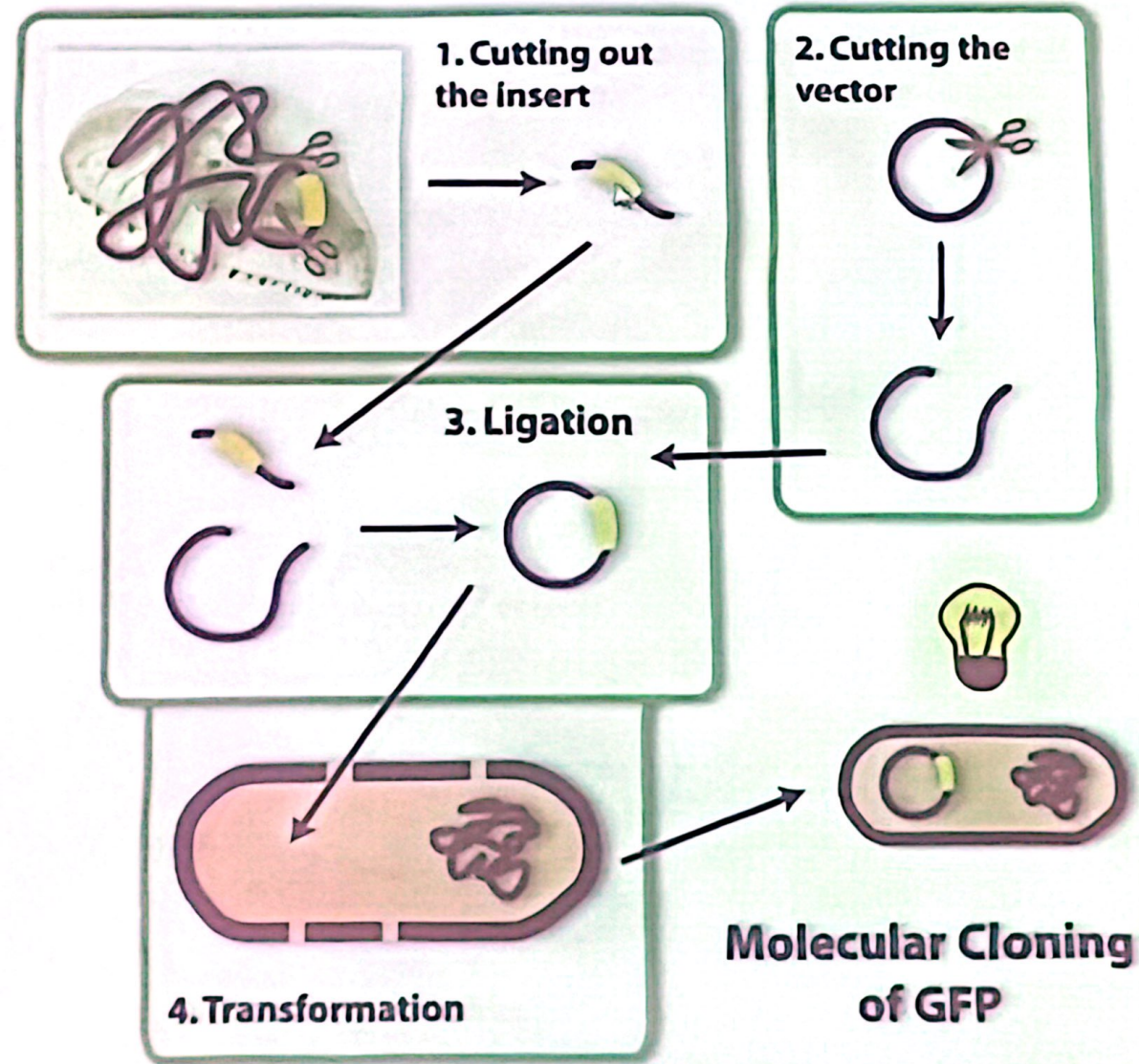
E. coli known as K12 serve as hosts for a wide range of vectors.

Steps to create recombinant DNA molecule in E. coli host cells

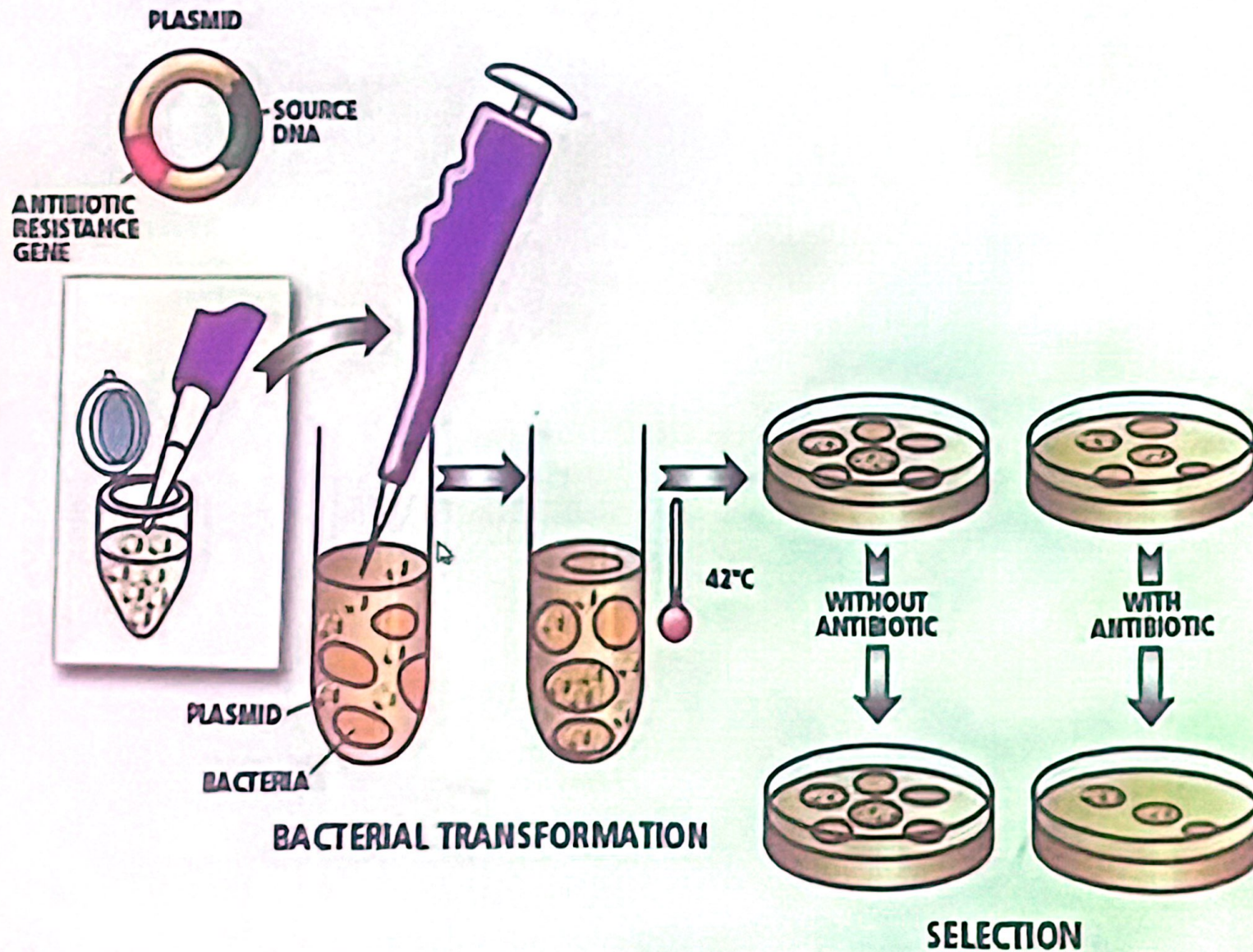
1. The DNA to be cloned is isolated and treated with restriction enzymes to create fragments ending in a specific sequence.

2. The fragments are ligated to plasmid molecules that have been cut with the same restriction enzyme, creating a recombinant vector molecule.

Steps of Recombinant DNA technology



Steps of Recombinant DNA technology



Steps of Recombinant DNA technology

3.The bacterial cells are heat-shocked to facilitate uptake of the recombinant vector into the cell.

4.The bacterial cells are cultured to allow the resident recombinant DNA molecules to replicate and form many copies.

5.The bacteria are grown on nutrient plates where they form colonies and are screened to identify those that have taken up recombinant plasmids.

Steps of Recombinant DNA technology

d. Production of Polypeptide from Cloned DNA:

One of the goals of genetic engineering is the production of a large quantity of a particular protein that is otherwise difficult to obtain.

In order to guarantee that a cloned gene can be transcribed into mRNA and then translated into polypeptide,
Certain requirements have to be fulfilled.

1. The cloned gene should not contain introns
2. Should be inserted in a place in the plasmid DNA following a promoter and a leader sequence.
3. It is easier to insert a foreign structural gene into the middle of a vector structural gene, which results in a hybrid polypeptide from which the foreign polypeptide can be released by appropriate chemical treatment.

Applications of Genetic Engineering

Recombinant DNA technology has made great contributions both in genetic research and in the application in the fields of medicine, veterinary medicine and agriculture:

a. **Characterization of gene structure by the application of DNA sequencing to cloned DNA.** 5... ATGGCTAACTATGGTAT.....3

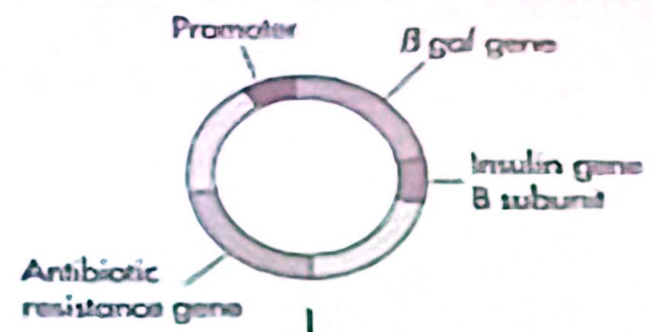
b. **Study of the expression and regulation of eukaryotic gene.**

c. **Processing Agricultural Products and Chemical:**

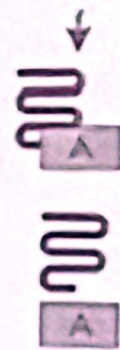
Genetically engineered bacteria and yeasts could be geared to manufacture commercially huge quantities of microbe-driven enzymes and chemicals (e.g. ethanol, acetone, etc).

d. **Production of Medically Important Proteins:**

Through recombinant DNA technology, it will become possible to produce important medical protein like human insulin, human growth hormone, interferon, ...etc.



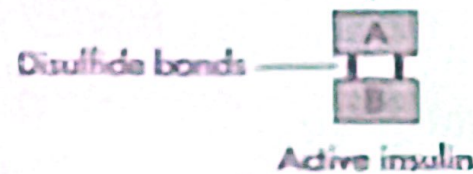
Transform into *E. coli* host



Extract and purify β -gal/insulin fusion proteins

Treat with cyanogen bromide to cleave A and B chains

Purify, mix A and B chains to form functional insulin



Applications of Genetic Engineering

e. Vaccine Production:

1. A vaccine against human hepatitis B

Vaccines made by genetic engineering offer the advantage that the microbial strains from which the proteins are extracted do not contain complete viruses, and thus there is no risk of accidental inoculation with live virus.

2. Foot-and-mouth (FMD)

is one of the most important animal diseases, and more FMD vaccine is produced throughout the world than any other vaccine.

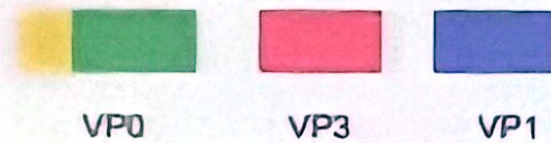
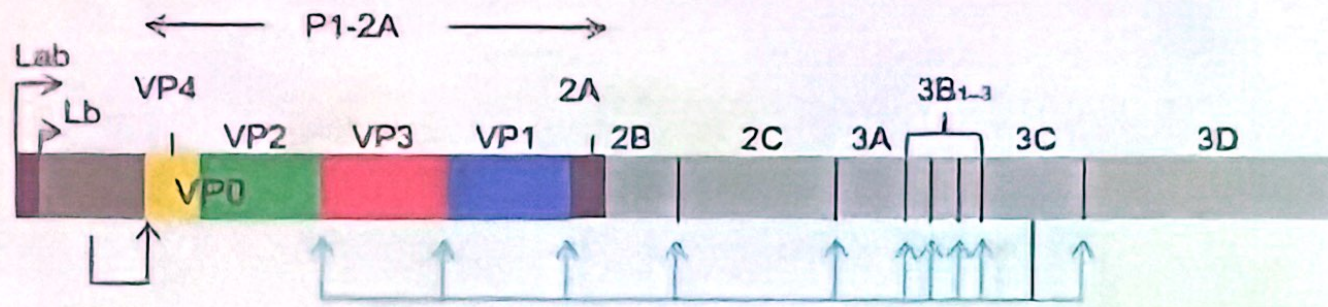
Applications of Genetic Engineering

The genome of FMD virus is a single strand RNA molecule, about 8000 bases (8 kb) long.

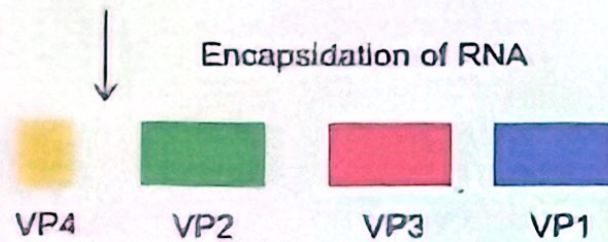
This RNA molecule codes, among other things, for four structural polypeptides, VP1, VP2, VP3 and VP4,

specify the protein coat of the virus.

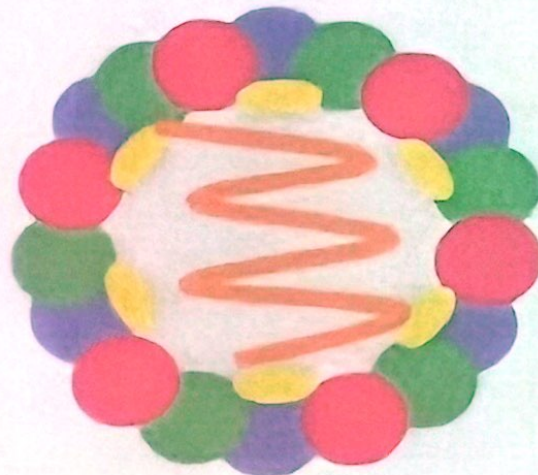
The antigenic portion of the virus is produced by VP1, which when inserted in plasmids and cloned in *E. coli*, could be transcribed and translated into a very reliable non-infective FMD vaccine.



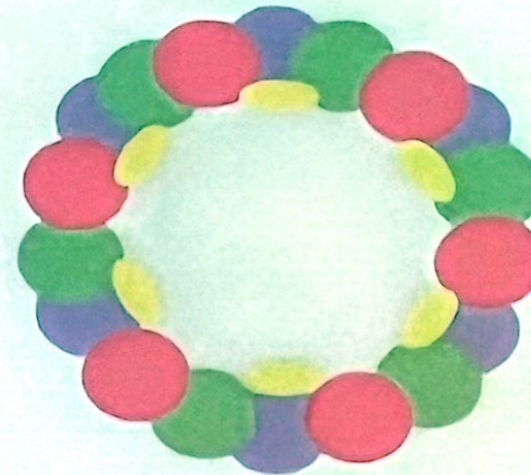
(Empty capsid particle proteins)



(Virus particle proteins)



Virus particle



Empty capsid particle